The Amber Fossil Fiasco

Whodunnit? · Linear Functions & Systems · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Linear Functions & Systems	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Writing equations of lines, linear models and scatter plots, systems of two and three equations, linear inequalities and linear programming.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Percival Hogg took it, in the Loading Dock, using this method: Tucked inside a plaster skull cast.

The reveal (read aloud): While the guests lined up for dessert, **Percival Hogg** slipped down to the **Loading Dock**, tucked the Golden Tear inside a **plaster skull cast**, and had it wheeled out as a "donation" for the cast collection. Every alibi and method you cleared pointed straight at him. Case closed, detective!

Clue map: what each correct clue clears

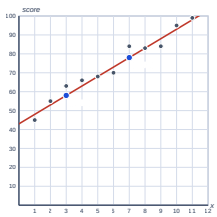
Card	Clears
1	Suspect: Odette Marsh · Place: Fossil Lab
2	Method: Hidden in a dinosaur-egg replica
3	Suspect: Rafferty Boone · Place: Sandbox Dig Site
4	Method: Swapped for a resin copy
5	Suspect: Imelda Sato · Place: Dinosaur Hall
6	Method: Folded into a souvenir T-shirt
7	Suspect: Cornelius Finch · Place: Gift Shop
8	Method: Packed in a field-supply crate
9	Suspect: Zuri Abara · Place: Ice Age Café
10	Method: Rolled up in a banner

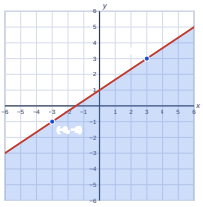
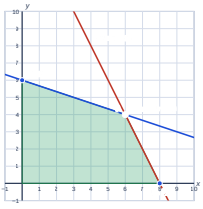
Answer Key

The Amber Fossil Fiasco · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution
1	Write the equation of the line through the two points. Any form is fine. $(-3, 4)$ and $(5, -2)$	**B.** $y = -\frac{3}{4}x + \frac{7}{4}$	Slope: $m = \frac{-2-4}{5-(-3}$ $-\frac{3}{4}$. Point-slope form with $-\frac{3}{4}(x+3)$. Distribu $x - \frac{9}{4}$. Add 4 to both sides: (Standard form: $3x +$
2	Write the equation of the line perpendicular to the given line that passes through the point $(6, -1)$. $3x - 5y = 20$	**D.** $y = -\frac{5}{3}x + 9$	Solve for y: $-5y = -$ $\frac{3}{5}x - 4$. The slope i Perpendicular slopes reciprocals: flip the fr change the sign, so m Point-slope through ($-\frac{5}{3}(x-6) = -\frac{5}{3}x$ $-\frac{5}{3}x + 9$. (Standard = 27.)
3	The museum's Fossil Camp charges a flat registration fee plus a fixed price for each night. The table shows the total cost for three stays. Find the cost of an 11-night stay.	**B.** \$425	Rate of change: $\frac{215}{5}$ = 35 dollars per nigh the next pair: $\frac{355-}{9-}$ 35, so the table is lin Flat fee: 110 = 35(2) The model is $C = 35n$ $C(11) = 35(11) + 40$ 425
4	The scatter plot shows hours of practice (x) and test score (y) for museum guide trainees. The line of best fit passes through the two labeled points. Use the line to predict the score of a trainee who practices 9 hours.	**D.** 88	Slope from the labele $\frac{78-58}{7-3} = \frac{20}{4} =$ hour. Point-slope with (3, 5 − 3), so $y = 5x + 43$ At $x = 9$: $y = 5(9) +$ predicted score is 88

Nights	2	5	9
Cost ($)	110	215	355



5	Solve the system. Write your answer as an ordered pair (a, c), where a is the price of an adult ticket and c is the price of a child ticket, both in dollars. Saturday: $2a + 3c = 55$ Sunday: $5a + 2c = 88$	**C.** (14, 9)	Eliminate c: multiply and Sunday by 3. Th $= 110$ and $15a + 6c$ Subtract: $11a = 154$, Back-substitute into $+ 3c = 55$, so $c = 9$. (14, 9): adult tickets child tickets cost \$9
6	Decide how many solutions the system has: one, none or infinitely many. $9x - 6y = 15$ $6x - 4y = 10$	**C.** infinitely many	Make the x-terms ma first equation by 2 ar 3. First: $18x - 12y = 30$ $12y = 30$. The two eq identical. Both equations descr line, so every point o infinitely many
7	Write the inequality for the graph. The boundary is a solid line through the two labeled points. Any form is fine. 	**A.** $y \le \frac{2}{3}x + 1$	Slope from the labele $\frac{3-(-1)}{3-(-3)} = \frac{4}{6} =$ crosses the y-axis at boundary is $y = \frac{2}{3}x$ The line is solid, so th includes the line ($\le$ c shading is below the Test (0, 0): $0 \le 1$ is tr in the shaded region $y \le \frac{2}{3}x + 1$ (equiva -3)
8	The museum workshop makes x skull casts and y egg kits each week. Mold time limits them to $2x + y \le 16$ and painting time to $x + 3y \le 18$, with $x \ge 0$ and $y \ge 0$. The profit is $P = 40x + 55y$ dollars. The feasible region is shown. Find the maximum weekly profit. 	**A.** \$460	The maximum is at a feasible region. The l are (8, 0) and (0, 6), origin. Find the missi solving $2x + y = 16$ a $y = 16 - 2x$, so $x +$ giving $x = 6$ and $y =$ Test every corner: $P($ $= 320$, $P(6, 4) = 240$ $P(0, 6) = 330$. The largest value is 4 the maximum profit i

9	The gift shop sells posters (p), mugs (m) and kits (k) at fixed prices. Three orders are shown. Find the price of each item as an ordered triple (p, m, k). Order A: $2p + m + k = 79$ Order B: $p + 3m + k = 107$ Order C: $p + m + 2k = 102$	**A.** (12, 20, 35)	Eliminate k twice. Or Order B: $p - 2m = -$ minus 2 times Order -56, so $3p + m = 56$ From $p = 2m - 28$: 3 $= 56$, so $7m = 140$ a $p = 12$. Back-substitute in Or $k = 79$, so $k = 35$. Th 20, 35)
10	Plan A costs \$85 plus \$12 per visit. Plan B costs \$35 plus \$17 per visit. Write an inequality for the number of visits x for which Plan A is the cheaper plan. Plan A cost < Plan B cost	**D.** $x > 10$	Write each cost: Plan and Plan B is $35 + 17$ cheaper when $85 +$ Subtract $12x$ and 35 $50 < 5x$. Divide by 5 inequality does not fl $x > 10$ (at exactly 10 cost the same, so 10

Wrong-Answer Map

The Amber Fossil Fiasco · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. $y = -\frac{3}{4}x + 4$	used the y-coordinate of the first point, 4, as the y-intercept, but the line crosses the y-axis at 7 over 4	The plaster skull casts were tapped one by one with a mallet. Every one rang solid, with nothing inside.
1	C. $y = -\frac{3}{4}x + \frac{25}{4}$	sign error in point-slope form: wrote $y - 4 = m(x - 3)$ instead of $y - 4 = m(x + 3)$	Odette Marsh was unloading a crate at the Loading Dock. The delivery driver has her signature at that exact minute.
1	D. $y = -\frac{4}{3}x$	slope found as run over rise: 8 over −6 instead of −6 over 8	Percival Hogg was admiring the Triceratops in the Dinosaur Hall. Three docents chatted with him the whole hour.
2	A. $y = \frac{3}{5}x - \frac{23}{5}$	used the slope of the given line (parallel), not the perpendicular slope	Percival Hogg was borrowing a magnifying glass in the Fossil Lab. The sign-out sheet has his name in blue ink.
2	B. $y = \frac{5}{3}x - 11$	took only the reciprocal of the slope and forgot to change the sign	Security opened the skull cast on the display stand. It was a hollow shell of plaster and nothing else.
2	C. $y = -\frac{3}{5}x + \frac{13}{5}$	took only the opposite of the slope, so the slope is not the negative reciprocal	Rafferty Boone was checking delivery trucks at the Loading Dock. The gate log has his initials on every entry.
3	A. $495	used the first total, 110, as the flat fee, but 110 already includes 2 nights	Every plaster cast in the education cart was tapped and shaken. Each was solid plaster, with no hidden hollow.
3	C. $605	divided the first total by 2 nights to get 55 per night, ignoring the flat fee	Imelda Sato was waiting for a courier at the Loading Dock. A security camera shows her there until the lights came up.
3	D. $385	multiplied the rate by 11 but left out the flat fee	Percival Hogg was buying a postcard at the Gift Shop. The receipt is stamped at the exact minute of the theft.
4	A. 45	used $y = 5x$ and left out the intercept 43	Percival Hogg was digging for plastic bones at the Sandbox Dig Site. Two parent volunteers sat beside him.
4	B. 103	added 5 times 9 to the score at 3 hours instead of using the change in x, $9 - 3$	Cornelius Finch was signing for a shipment of mugs at the Loading Dock. The shipping clerk wrote down the time.
4	C. 70	found the slope as 20 over 10 by adding the x-values instead of subtracting	A volunteer X-rayed the skull casts at the museum clinic. The images showed plaster all the way through.
5	A. (8, 13)	this pair works in the Saturday equation only; it fails 5a + 2c = 88	Zuri Abara was leading a late tour group through the Loading Dock. Eleven guests have her in their photos.
5	B. (16, 4)	this pair works in the Sunday equation only; it fails 2a + 3c = 55	The cast of the Velociraptor skull was cut in half at the workshop. It was solid plaster, with no cavity.

5	D. (9, 14)	swapped the answers: a is the adult price, c is the child price	Percival Hogg was ordering a latte at the Ice Age Café. The barista wrote his name on the cup and the time on the ticket.
6	A. none	treated the identical equations like a contradiction such as 0 = 5, but here the result is 0 = 0, which is always true	Percival Hogg was reading the labels in the Dinosaur Hall. A tour group of nine says he stood with them the entire hour.
6	B. exactly two	two lines can meet at most once; these are the same line, which meets everywhere	The plaster skull casts were carried to the lab and checked. Every one was solid, with no hollow space at all.
6	D. exactly one	assumed two different-looking equations must cross at one point, but both have slope 3 over 2 and intercept −5 over 2	Rafferty Boone was walking the Loading Dock with his flashlight. Two truck drivers watched him make the round.
7	B. $y \le \frac{3}{2}x + 1$	slope found as run over rise: 6 over 4 instead of 4 over 6	A curator tapped the mammoth skull cast with a hammer. It was solid plaster, and nothing was tucked inside.
7	C. $y < \frac{2}{3}x + 1$	used a dashed-line symbol: the line is solid, so the inequality must include equal to (≤)	Odette Marsh was inspecting a shipment of bones at the Loading Dock. The receiving form is signed in her handwriting.
7	D. $y \ge \frac{2}{3}x + 1$	shaded the wrong side: the region is below the line, so it needs ≤, not ≥	Percival Hogg was talking with the lab staff in the Fossil Lab. The lab supervisor jotted down every word.
8	B. $490	swapped the coordinates of the missing corner, using (4, 6); that point is outside the region because 4 + 3(6) = 22 is more than 18	Imelda Sato was sorting packing foam at the Loading Dock. A coworker says they worked side by side.
8	C. $650	used the largest x and the largest y separately, (8, 6), which is not a corner of the region	The set of skull casts for the school program was opened up. Every one was just a plain plaster shell.
8	D. $330	tested only the two corners that were labeled on the graph, (8, 0) and (0, 6), and missed the corner (6, 4)	Percival Hogg was choosing a souvenir mug at the Gift Shop. The shop camera shows him at the counter the whole time.
9	B. (22, 18, 31)	fits Orders B and C but not Order A: the first equation was never checked	Cornelius Finch was restocking pallets at the Loading Dock. The forklift log has his name at that exact minute.
9	C. (16, 22, 25)	fits Orders A and B but not Order C: the third equation was never checked	Percival Hogg was building a sandcastle with children at the Sandbox Dig Site. The photographer has him in every picture.
9	D. (12, 35, 20)	the right three prices in the wrong order: the middle value is the mug and the last is the kit	Police searched the cast storage room. Every plaster skull was solid and labeled, and none had been touched.
10	A. $x < 10$	divided by −5 without flipping the inequality symbol	Percival Hogg was sipping cocoa at the Ice Age Café. The cashier remembers because he left a very large tip.
10	B. $x > -10$	sign slip when moving the constants: 85 − 35 was written as 35 − 85, giving $-5x < 50$	The skull casts were scanned with a handheld detector at the workshop. None of them showed anything but plaster.
10	C. $x \ge 10$	included 10, but at 10 visits the plans cost the same, so Plan A is not cheaper	Zuri Abara was loading the tour bus at the Loading Dock. The bus driver signed her departure sheet.